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The nanogranular origin of friction and cohesion in shale—A strength homogenization approach to interpretation of nanoindentation results

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SUMMARY

An inverse micromechanics approach allows interpretation of nanoindentation results to deliver cohesive-frictional strength behavior of the porous clay binder phase in shale. A recently developed strength homogenization model, using the Linear Comparison Composite approach, considers porous clay as a granular material with a cohesive-frictional solid phase. This strength homogenization model is employed in a Limit Analysis Solver to study indentation hardness responses and develop scaling relationships for indentation hardness with clay packing density. Using an inverse approach for nanoindentation on a variety of shale materials gives estimates of packing density distributions within each shale and demonstrates that there exists shale-independent scaling relations of the cohesion and of the friction coefficient that vary with clay packing density. It is observed that the friction coefficient, which may be interpreted as a degree of pressure-sensitivity in strength, tends to zero as clay packing density increases to one. In contrast, cohesion reaches its highest value as clay packing density increases to one. The physical origins of these phenomena are discussed, and related to fractal packing of these nanogranular materials. Copyright © 2010 John Wiley & Sons, Ltd.

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KEY WORDS: nanoindentation; shale; cohesive-frictional behavior; strength homogenization; poromechanics; fractal packing

1. INTRODUCTION

Nanoindentation is one of the most promising techniques that has emerged from the implementation of nanotechnology in material science and engineering for assessing mechanical properties at small scales. The idea is simple: by pushing a needle onto the surface of a material, the surface deforms in a way that reflects the mechanical properties of the indented bulk material. Yet, in contrast to most metals and ceramics, for which this technique was originally developed, most materials relevant to civil engineering, petroleum engineering or geophysical applications, are highly heterogeneous

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from a scale of a few nanometers to macroscopic scales. The most prominent heterogeneity is the porosity.

Shale, for instance, is a multi-phase, multi-scale, and compositionally diverse class of materials that is particularly critical for the petroleum industry [1]. Strength of shales is particularly important because borehole stability problems, an important issue during petroleum drilling operations, can be viewed as strength failure problems. The complex and diverse nature of these materials, however, currently demands a detailed laboratory testing of localized samples to assess strength behavior. An improved understanding of the fundamental strength behavior of shale can lead to predictive models that circumvent costly and time-consuming experiments. For this purpose, it is convenient to dissect the complex mechanical behavior of shale into different scales, as shown in Figure 1. Structured around a three-level 'thought model' of the multi-scale structure of shale materials [2, 4, 5], this thought model spans roughly seven orders of magnitude, from the scale of the elementary particle (Level '0') to the scale of the macroscopic clay–silt inclusion composite (Level 'II'). The different scales satisfy the scale separability condition; that is the characteristic length scale of Level '0' is much smaller than the characteristic length scale of Level 'I' which is much smaller than the characteristic length scale of Level 'II'. In some more detail:

- Level 'II' is the macroscopic (lab bench) scale of 'intact' rock samples of characteristic size in the sub-millimeter to millimeter range. At this scale, the material is composed of a porous clay composite intermixed with an abundant population of poorly sorted detrital grains (quartz or feldspar inclusions). The volume fraction of this silt phase is denoted by f_{inc}. As regards strength, this Level 'II' has been the focus of the classical strength of materials tests, such as unconfined compression tests [6], triaxial tests [7], or shear tests [8]. Several attempts have been made at this scale to fit the macroscopic experimental data to some predefined strength criteria, such as Mohr–Coulomb, Drucker–Prager, or combinations thereto, such as the modified Lade criterion (e.g. [9]).
- Level 'I' is the scale of the porous clay composite which manifests itself at the sub-micrometer scale. The morphology of this scale may be observed by advanced methods, such as scanning electron microscopy (SEM). SEM images reveal the varying configurations of the porous clay composite, ranging from highly ordered sheet bundles to wavy flake structures, with characteristic dimensions of the clay structures of 500–1000 nm wide and 20–50 nm thick [4]. In contrast to these visual observations, however, Bobko and Ulm used observations from nanoindentation measurements to interpret the mechanical morphology at Level 'I' as a composite of 'mechanically acting' spherical particles [5]. Mercury intrusion porosimetry (MIP) testing has determined that the space in between these bundles and flakes of clay structures account for almost the totality of the porosity in shale, with characteristic pore access radii of some nanometers. The packing density of the porous clay is denoted by η, with the clay porosity, $\varphi = 1 - \eta$. The packing density at Level 'I' may be calculated from macroscopically observed porosity, n, and inclusion volume fraction, f_{inc}, by:

$$\eta = 1 - \frac{n}{1 - f_{\text{inc}}} \tag{1}$$

  Analytical strength models for porous clay materials originate from the Cam Clay-type models for clay soil behavior [10, 11]. More recently, discrete element simulations [12], granular physics investigations [13] and strength homogenization approaches [14–22] have all made important contributions to the topic; leading to the current understanding that a cohesive-frictional solid intermixed with some porosity follows—in first order (that is for low and medium confinement levels)—an elliptical strength criterion. On the other hand, the implementation of such advanced micromechanics approaches requires as input the solid strength properties, e.g. solid cohesion and friction coefficient of the clay solid phase.
- Level '0' is the scale of agglomerations of elementary clay unit structures. These elementary clay unit structures, such as illite, smectite, and kaolinite, have been classically designated as the fundamental scale of clay mineralogy (see e.g. [23, 24]). In accordance with contemporary clay nomenclature, contained within these 'clay unit structures' are 'layers' containing one or

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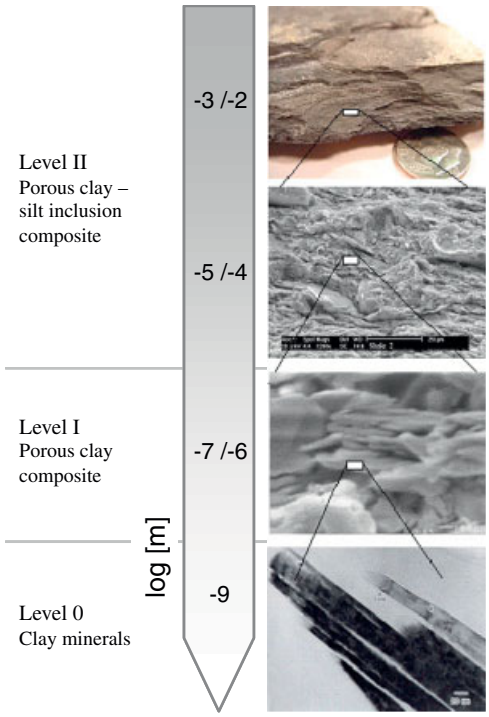


Figure 1. Multi-scale structure thought model of shale (adapted from [2]). Level 0 corresponds to the scale of the elementary clay particles that form the solid clay phase in shale. Level I is the scale of the solid clay particles and the (nano)porosity that forms the porous clay composite. Level II is the macroscopic scale of characteristic length scale in the sub-millimeter and millimeter range, at which shale is composed of the porous clay fabric and silt-size inclusions (mostly quartz). Level II and level I images were obtained from the scanning electron microscope (SEM) imaging. SEM micrographs have shown various porous clay configurations—from highly ordered bundles of platelets to wavy flake structures, with clay sheets of characteristic dimensions of 500–1000 nm and 20–50 nm thick. Porosimetry testing has established the characteristic pore access radii of some nanometers, which assigns most of shale's porosity to the level I length scale. The level 0 image was obtained from the transmission electron microscope (TEM) imaging [3, with the permission of Springer Science and Business Media].

more tetrahedral sheets and an octahedral sheet of aluminosilicates, and 'inter-layer material' which may consist of cations, hydrated cations, organic materials, hydroxide octahedra, and/or hydroxide octahedral sheets [25]. While the structures of the individual clay sheets and layers are reasonably well known, their mechanical properties, and the mechanical properties of clay unit structures, are rarely documented in handbooks [26]. The small size of clay unit structures in a pure solid form, even when agglomerated, is the main obstacle for performing direct measurements of their mechanical properties. Recent research has employed analysis of nanoindentation experiments [5] and back analysis of UPV measurements [2, 27] to identify the shale invariant elastic stiffness properties of Level '0'. These stiffness properties, which do not vary between shale materials, are transversely isotropic. By considering a spherical particle shape, all elastic anisotropies manifested at Level 'I' are assumed to be intrinsic to Level '0.' Such an understanding typed into a quantitative model for compacted clay as present in shale is still missing for strength properties.

The overall goal of this work is to quantify the strength properties of compacted clays in shale, which may serve as input or provide guidance for the prediction of the strength behavior of shales. To reach our goal we implement a recently obtained strength homogenization model for porous composites with a cohesive-frictional solid phase [28] into a computational limit

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analysis environment [29, 30] for the derivation of indentation hardness–constituent strength properties–microstructure relations. The approach refines previous developments of hardness–packing density relations for cohesive-frictional porous materials [21] and its application in statistical indentation analysis [31]. More precisely, the scaling functions derived here extend the domain of application of previous approaches which were limited to small solid friction angles. This extension makes the scaling functions a formidable tool to sense microstructure of cohesive-frictional porous solids by matching a distribution of experimentally measured indentation hardness results to a distribution of packing densities and to mechanical properties of the solid. Finally, when this inverse approach is applied to the results of a massive indentation campaign on shale materials, new observations and insights into the origin of the cohesive-frictional strength behavior of shale are made.

2. MATERIALS AND METHOD

To motivate the forthcoming developments, consider an indentation test into a porous material. The characteristic size of the porosity is much smaller than the maximum indentation depth such that scale separability applies. The indentation test therefore senses a composite response of the solid phase and the porosity. That is, the hardness H and the indentation modulus M determined from the load–depth ($P-h$) curve (Figure 2) according to classical indentation analysis tools [32], are representative of the particle properties (particle stiffness m_s, hardness h_s, friction coefficient μ, etc.), the solid concentration or packing of particles η, as defined by (1), and some morphological parameters (denoted by η_0); that is:

$$M = \frac{\sqrt{\pi}}{2}\frac{S}{\sqrt{A_c}} = m_s \times \Pi_M\left(\frac{\mathbb{C}^s}{m_s}, \eta, \eta_0\right) \tag{2}$$

$$H = \frac{P_{\max}}{A_c} = h_s \times \Pi_H(\mu, \eta, \eta_0) \tag{3}$$

where $S = \mathrm{d}P/\mathrm{d}h$ is the initial slope of the unloading branch of the $P-h$ curve, $P_{\max}$ is the maximum indentation load, and A_c is the projected contact area, expressed as a function of the maximum indentation depth, $h_{\max}$ as calculated by the Oliver and Pharr method [32]; while Π_M and Π_H are dimensionless scaling relations pertaining to stiffness and hardness, respectively. Linear and non-linear microporomechanics theory provides a suitable framework for determining these scaling relations.

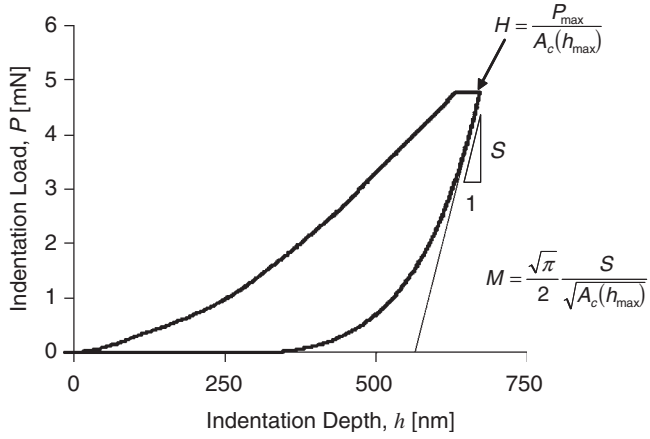


Figure 2. A typical load versus depth curve for nanoindentation on the porous clay in shale. The indentation hardness, H, and indentation modulus, M, are derived from this curve.

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2.1. Linear microporomechanics

In the case of a linear elastic behavior of the solid phase, linear micromechanics theory provides the following estimate of the drained stiffness tensor of the solid-pore composite with similar inclusion morphologies within the framework of the self-consistent scheme (see, for instance, [19, 33]):

$$\mathbb{C}^{\hom} = \eta \mathbb{C}^s : [\mathbb{I} + \mathbb{P}^{SC} : (\mathbb{C}^s - \mathbb{C}^{\hom})]^{-1} \tag{4}$$

where $\mathbb{C}^s$ is the stiffness tensor of the solid phase, $\mathbb{I}$ is the fourth-order unit tensor, while $\mathbb{P}^{SC}$ is the Hill tensor characterizing particle interactions, particle shape etc. Implemented for shale elastic anisotropy, it was recently found that among various combinations of different sources of anisotropy (particle shape and orientation, intrinsic particle anisotropy), the elastic response of the clay fabric represented as a granular ensemble of aligned effective clay particles with spherical morphology and transversal isotropic elasticity matches best the elasticity content sensed by nanoindentation and ultrasonic pulse velocity measurements at nano- and macroscopic length scales, respectively [27, 34]. The solid stiffness components do not vary between shale materials, with stiffness components in Voigt notation: $C^s_{11} = 44.9\,\text{GPa}$, $C^s_{12} = 21.7\,\text{GPa}$, $C^s_{13} = 18.1\,\text{GPa}$, $C^s_{33} = 24.2\,\text{GPa}$, and $C^s_{44} = 3.7\,\text{GPa}$. Stiffness expression (4) serves as input for the determination of the indentation moduli, which for transversely isotropic materials, depends on the direction of the indentation relative to the axis of material symmetry (noted $M^{(3)}$) and parallel to the axis of symmetry (noted $M^{(1)}$) [35]:

$$M^{(3)} = 2\sqrt{\frac{C_{11}C_{33} - C_{13}^2}{C_{11}}\left(\frac{1}{C_{44}} + \frac{2}{\sqrt{C_{11}C_{33}} + C_{13}}\right)^{-1}} \tag{5a}$$

$$M^{(1)} \approx \sqrt{\sqrt{\frac{C_{11}}{C_{33}}}\frac{C_{11}^2 - C_{12}^2}{C_{11}} M_3} \tag{5b}$$

Given the shale invariant nature of the solid stiffness properties, the asymptotic (solid) values for the indentation modulus are $m_{\text{s}}^{(3)} = \lim_{\eta \to 1} M^{(3)} = 14.8\,\text{GPa}$ and $m_{\text{s}}^{(1)} = \lim_{\eta \to 1} M^{(1)} = 26.3\,\text{GPa}$, displaying a strong anisotropy of the level '0' particle. A combination of the previous equations thus provides a means to determine the dimensionless scaling relations of the form (2) for the two indentation directions. Fitted to polynomial expression, the following continuous relations were obtained [31]:

$$\frac{M^{(i)}}{m_{\text{s}}^{(i)}} = \Pi_M^{(i)} = \begin{cases} 1.28\langle\eta - \eta_0\rangle + 3.09\langle\eta - \eta_0\rangle^2 - 5.05\langle\eta - \eta_0\rangle^3 + 3.54\langle\eta - \eta_0\rangle^4, & i = 1 \\ 2.10\langle\eta - \eta_0\rangle - 1.15\langle\eta - \eta_0\rangle^2 + 3.14\langle\eta - \eta_0\rangle^3 - 2.43\langle\eta - \eta_0\rangle^4, & i = 3 \end{cases} \tag{6}$$

where $\eta_0 = 0.5$ is the percolation threshold pertaining to a spherical particle geometry or highly aligned particles (cf. [34]), while $\langle x\rangle = x$ for $x > 0$, and $\langle x\rangle = 0$ for $x < 0$.

2.2. Non-linear microporomechanics

The upscaling of strength properties is more involving requiring the use of non-linear homogenization theory. Within the context of yield design theory, the problem comes to solve for a representative volume element (*rev*) (for details see, for instance, [19]):

$$\pi^{\hom} = \inf_{\underline{v}^* \in U(\mathbf{D})} \overline{\pi(\underline{z}, \mathbf{d}(\underline{v}^*))}, \quad U(\mathbf{D}) = \{\underline{v}^* | \underline{v}^* = \mathbf{D} \cdot \underline{z}\} \tag{7}$$

where $\pi(\mathbf{d}) = \sup(\sigma_{ij} d_{ij})$ stands for the local dissipation function, $\mathbf{D} = \overline{\mathbf{d}(\underline{z})}$ is the macroscopic strain rate tensor, which is the volume average (indicated by an overbar) of the local strain rates $\mathbf{d}(\underline{z})$ that derive from kinematically admissible (KA) velocity fields $\underline{v}^*$ satisfying the regular strain rate boundary condition stated in (7). An elegant way of estimating the macroscopic dissipation function $\pi^{\hom}$ is the Linear Comparison Composite approach suggested by Ponte Castañeda [36, 37].

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Implemented for the pore-solid composite [28], the approach comes to estimating π^{hom} from the stationary condition

$$\widetilde{\pi}^{\text{hom}}(\mathbf{D}) = \underset{\mathbb{C}^s,\boldsymbol{\tau}}{\text{stat}}[\Psi(\mathbf{D}) + \eta\Upsilon^s] \tag{8}$$

where $\Psi(\mathbf{D})$ is the strain rate energy function of the composite obtained from linear homogenization theory; while function Υ^s (assumed constant in each phase, and zero in the pore space) measures the non-linearity of the composite material as the difference between the solid's dissipation and the strain rate energy; that is:

$$\Psi(\mathbf{D}) = \eta\overline{\psi^s(\mathbf{d})}^{V_s} = \eta \inf_{\underline{v}^* \in \mathscr{K}(\mathbf{D})} \overline{\frac{1}{2}\mathbf{d}:\mathbb{C}^s:\mathbf{d} + \boldsymbol{\tau}:\mathbf{d}}^{V_s} \tag{9}$$

$$\Upsilon^s = \underset{\mathbf{d}}{\text{stat}}\{\pi^s(\mathbf{d}) - \psi^s(\mathbf{d})\} \tag{10}$$

where the solid stiffness of the comparison composite $\mathbb{C}^s$ and the prestress $\boldsymbol{\tau}$ serve as optimization parameters to enforce the stationarity conditions. Specification of the previous relations for a solid whose strength domain is described by a Drucker–Prager criterion (solid cohesion c_s, friction coefficient α) yields for a solid-pore composite [28]

$$\widetilde{\pi}^{\text{hom}}(D_v, D_d) = \Sigma_0 D_v - \text{sgn}(2\alpha^2\mathscr{K} - \eta)\sqrt{\mathscr{A}D_v^2 + 4\mathscr{B}D_d^2} \tag{11}$$

$$\text{subjected to: } (\mathscr{A}D_v^2 + 4\mathscr{B}D_d^2) \geqslant 0$$

where $D_v = \text{tr}\,\mathbf{D}$ and $D_d = \sqrt{\frac{1}{2}\boldsymbol{\Delta}:\boldsymbol{\Delta}}$ are the macroscopic strain rate invariants, with $\boldsymbol{\Delta} = \mathbf{D} - \frac{1}{3}D_v\mathbf{1}$; while Σ_0, $\mathscr{A}$, and $\mathscr{B}$ for the Drucker–Prager solid phase read as:

$$\frac{\mathscr{A}}{(c_s)^2} = \frac{\eta^2\mathscr{K}(\eta - \alpha^2\mathscr{K})}{(\eta - 2\alpha^2\mathscr{K})^2}, \quad \frac{\mathscr{B}}{(c_s)^2} = \frac{\eta\mathscr{M}(\eta - \alpha^2\mathscr{K})}{\eta - 2\alpha^2\mathscr{K}}, \quad \frac{\Sigma_0}{c_s} = \frac{\eta\alpha\mathscr{K}}{2\alpha^2\mathscr{K} - \eta} \tag{12}$$

Herein, $\mathscr{K}$ and $\mathscr{M}$ are dimensionless pore morphology factors that stem from the use of a specific linear upscaling scheme of the bulk and shear modulus in which the solid's shear-to-bulk modulus ratio is replaced by the Drucker–Prager friction coefficient of the solid phase:

$$\mathscr{K} = \frac{K^{\text{hom}}}{g^s} = \mathscr{K}\left(\frac{g^s}{k^s} \to \alpha^2, \eta, \eta_0\right), \quad \mathscr{M} = \frac{G^{\text{hom}}}{g^s} = \mathscr{M}\left(\frac{g^s}{k^s} \to \alpha^2, \eta, \eta_0\right) \tag{13}$$

Two schemes for obtaining morpholgy factors are used frequently [38] based on the microstructure of the material:

- For a matrix—pore inclusion morphology, the Mori–Tanaka scheme provides the following estimate for $\mathscr{K}$ and $\mathscr{M}$:

$$\mathscr{K}_{\text{MT}} = \frac{4\eta}{3(1-\eta) + 4\alpha^2}, \quad \mathscr{M}_{\text{MT}} = \frac{\eta(9 + 8\alpha^2)}{15 - 6\eta + (20 - 12\eta)\alpha^2} \tag{14}$$

- For a perfectly disordered material, in which none of the mechanical phases plays a specific role of matrix or inclusion, and which resembles granular or polycrystal morphologies, the self-consistent scheme provides the following estimates:

$$\begin{aligned}
\mathscr{K}_{\text{SC}} &= \frac{4\eta\mathscr{M}_{\text{SC}}}{4\alpha^2\mathscr{M}_{\text{SC}} + 3(1-\eta)} \\
\mathscr{M}_{\text{SC}} &= \frac{1}{2} - \frac{5}{4}(1-\eta) - \frac{3}{16\alpha^2}(2+\eta) \\
&\quad + \frac{1}{16\alpha^2}\sqrt{144(\alpha^4 - \alpha^2) - 480\alpha^4\eta + 400\alpha^4\eta^2 + 408\alpha^2\eta - 120\alpha^2\eta^2 + 9(2+\eta)^2}
\end{aligned} \tag{15}$$

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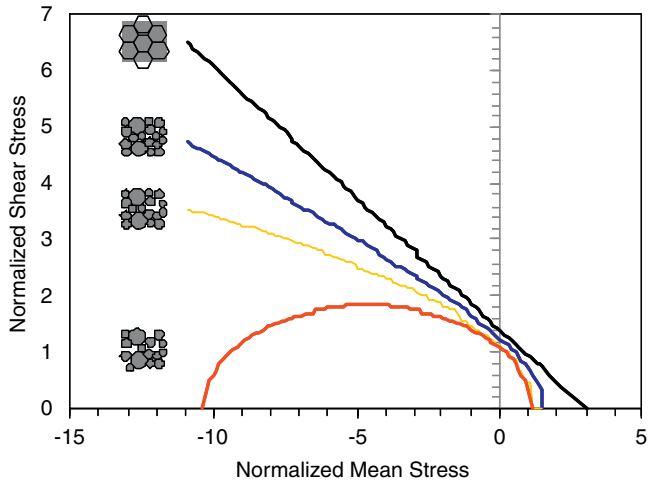


Figure 3. Strength domain of a cohesive frictional solid with packing density.

Note that $\mathscr{K}_{\rm SC}=\mathscr{M}_{\rm SC}=0$ at the percolation threshold $\eta_0=1/2$. This percolation threshold is consistent with the random loose packing fraction of spheres [39], making the SC scheme a popular choice for the micromechanical modeling of granular materials.

Finally, we make use of the yield design definition that links stresses to the dissipation function:

$$\boldsymbol{\Sigma}=\frac{\partial\widetilde{\pi}^{\rm hom}}{\partial\mathbf{D}},\quad \Sigma_m=\frac{1}{3}{\rm tr}\boldsymbol{\Sigma}=\frac{\partial\widetilde{\pi}^{\rm hom}}{\partial D_v},\quad \boldsymbol{\Sigma}_d=\sqrt{\frac{1}{2}\mathbf{S}:\mathbf{S}}=\frac{1}{2}\frac{\partial\widetilde{\pi}^{\rm hom}}{\partial D_d} \tag{16}$$

where $\mathbf{S}=\boldsymbol{\Sigma}-\Sigma_m\mathbf{1}$. The application of (16) to (11) yields the sought homogenized strength criterion:

$$F(\Sigma_m,\Sigma_d)={\rm sign}(\mathscr{B})\left(\frac{(\Sigma_m-\Sigma_0)^2}{\mathscr{A}}+\frac{\Sigma_d^2}{\mathscr{B}}\right)-1\leqslant 0 \tag{17}$$

It is recognized that the sign of $\mathscr{B}$ determines whether the homogenized strength criterion is an elliptical criterion or a hyperbolic criterion, as displayed in Figure 3. For the Drucker–Prager solid, there exists a critical packing density for which the strength domain is open (hyperbolic case) or closed (elliptical case) on the hydrostatic axis. This critical packing density depends on the homogenization scheme (MT or SC):

$$0\leqslant\eta_{\rm crit}^{\rm MT}=1-\frac{4}{3}\alpha^2\leqslant 1 \tag{18}$$

$$\frac{2}{3}\leqslant\eta_{\rm crit}^{\rm SC}=1-\frac{\sqrt{1216\alpha^4+432\alpha^2+81}-(16\alpha^2+9)}{2(20\alpha^2+3)}\leqslant 1 \tag{19}$$

Note that the present strength homogenization model differs somewhat from the original strength homogenization models developed by Dormieux and co-workers [16, 19], within the framework of the effective strain rate approach. The earlier approach restricts the domain of application to small solid friction coefficients [21], relevant for the elliptical strength domain. This restriction has been removed here through the application of the Linear Comparison Composite Approach, which opens the way to considering the transition from the elliptical to the hyperbolic case, for a larger range of solid friction coefficients, defined by the domain of application of the Drucker–Prager model [40]:

$$0\leqslant\alpha<\sqrt{\frac{3}{4}} \tag{20}$$

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On the other hand, as discussed in [19, 21], and shown recently by a comparison with numerical hollow sphere computational yield design solutions in [22], it is expected that the quadratic form of the strength criterion (17) will limit its accuracy to moderate hydrostatic pressures, and that it will perform poorly close and around hydrostatic compression stress states. While a shortcoming of the model, this is not a concern for the intended use of the model for indentation hardness. The stress field in an indentation test—at yield—is not hydrostatic, but shear dominated, activating the shear strength capacity of the indented porous material at low to moderate hydrostatic confinement states [21].

2.3. Implementation for hardness analysis

Empirical relationships between indentation hardness and material strength properties of solids were pioneered by Brinell [41], and many subsequent contributions have developed mechanics solutions for indentation in cohesive solids (e.g. [42–44]) and cohesive-frictional solids [45]. We are now ready to implement the strength homogenization result (11) and (17) for hardness analysis. Consider thus an indenter at depth h. Within the limit theorems of yield design, we seek for a link of the form [21, 45]:

$$H^- \leqslant \frac{1}{A_c} \sup_{F(\boldsymbol{\Sigma}^* \mathrm{SA}) \leqslant 0} \int_\Omega \Sigma^*_{ij} D_{ij} \, \mathrm{d}\Omega = \frac{1}{A_c} \inf_{\underline{U}^* \mathrm{KA}} \int_\Omega \widetilde{\pi}^{\mathrm{hom}}(D_{ij}(\underline{U}^*)) \mathrm{d}\Omega \leqslant H^+ \tag{21}$$

where the lower bound is based on stress fields Σ^*_{ij} that a statically admissible (SA) and compatible with the strength criterion (17), while the upper bound is based on KA velocity fields $\underline{U}^*$ evaluated for a unit indentation rate, and an evaluation of the dissipation capacity of the porous material as captured by the homogenized support function (11). Cariou *et al.* [21] pioneered the use of the upper bound for the derivation of hardness–packing density scaling relations for cohesive frictional solids; but the approach is restricted to the case of elliptical strength domains for the porous composite, which prevail only for certain combinations of friction and packing density. This work aims to remove or relax the restrictions on friction, cohesion, and packing density by using the homogenized strength domain that considers the additional possibility of hyperbolic strength domains for the porous composite. Contemporary computational limit analysis solvers [29, 30] can accommodate the yield design problem. More specifically, to use the Limit Analysis Solver, a model material domain is discretized by a finite element mesh. This axisymmetric mesh is loaded by a conical indenter with equivalent cone angle to a Berkovich tip ($\theta = 70.32^\circ$). The solver employs both stresses and velocities as degrees of freedom, requiring that the stress fields satisfy the weak form of equilibrium and the chosen strength criterion, and that the velocity fields are KA and obey an associated flow rule. The principle of limit analysis is to directly find a numerical estimate of the stress and velocity fields which obey these conditions of plasticity.

The limit analysis solver was used to predict indentation hardnesses, normalized by solid cohesion, as a function of varying solid friction coefficient and packing density. Numerical convergence was obtained for all values of packing density and all values of solid friction coefficient $\alpha < 1/\sqrt{3}$. Figure 4 presents a sampling of results in terms of relationships between packing density and hardness to cohesion ratios for various friction coefficients, α, for a self-consistent solid-pore morphology. Comparison with results of Cariou *et al.* [21], also presented in Figure 4, shows good agreement (where there are any discrepancies, Cariou's solutions prove to be an upper bound, as expected), despite the fact that the two approaches employ differently derived homogenized strength criteria and different methods of solving for the stress and velocity fields. The comparison demonstrates that the strength homogenization approach considered here, even with the limitation on values of $\alpha > 1/\sqrt{3}$, allows for a larger range of friction coefficients than Cariou *et al.*'s approach.

Continuous fitting functions that summarize the discrete simulation results in closed-form expressions that span the entire range from the elliptical to the hyperbolic case are useful for data analysis.

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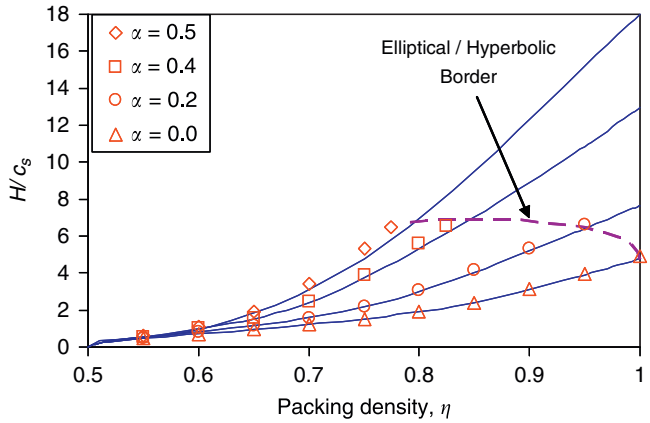


Figure 4. Relationships between hardness to cohesion ratios, H/c_s and packing density, η, for solid friction coefficients of $\alpha = 0$, 0.2, 0.4, and 0.5. Solid lines are the scaling results developed through use of the limit analysis solver. Points are data from [21], which employed an upper bound in conjunction with the effective strain rate approach. The improvement over Cariou *et al.*'s approach is evident, as elliptical strength domains do not apply to the entire parameter space.

The format of the scaling relations is chosen in the form (3):

$$\frac{H}{h_s(c_s,\alpha)} = \Pi_H(\alpha,\eta,\eta_0) \tag{22}$$

where $\Pi_H(\alpha,\eta,\eta_0)$ is a dimensionless function to be developed, and $h_s = h_s(c_s,\alpha) = \lim_{\eta\to 1} H$ is the asymptotic hardness of a cohesive-frictional solid phase that obeys the Drucker–Prager criterion. This asymptotic value does not depend on the employed linear homogenization scheme, but relates only to the solid's cohesion, c_s, and friction coefficient, α, by the fitted function

$$h_s = c_s \times a(1+b\alpha+(c\alpha)^3+(d\alpha)^{10}) \tag{23}$$

with fitted parameters:

$$a = 4.7644$$

$$b = 2.5934$$

$$c = 2.1860$$

$$d = 1.6777$$

The dimensionless function $\Pi_H \in [0,1]$ depends on the packing density η and the solid's friction coefficient α. Analogously to the work of Cariou *et al.* [21], the functional form is based on a limit development around α:

$$\Pi_H(\alpha,\eta) = \Pi_1(\eta,\eta_0) + \alpha(1-\eta)\ \Pi_2(\alpha,\eta,\eta_0) \tag{24}$$

where the first part of the function is independent of the friction coefficient. For completeness, dimensionless functions $\Pi_1(\eta,\eta_0)$ and $\Pi_2(\alpha,\eta,\eta_0)$ are fitted for the two considered pore morphologies: the Mori–Tanaka scheme ($\eta_0 = 0$) and the self-consistent scheme ($\eta_0 = 1/2$):

- For the Mori–Tanaka scheme, which corresponds to a matrix-pore morphology, we obtain the following expressions for $\Pi_1(\eta,\eta_0 = 0)$ and $\Pi_2(\alpha,\eta,\eta_0 = 0)$:

$$\begin{aligned} \Pi_1^{\mathrm{MT}}(\eta) &= \eta(1+g(1-\eta)+h(1-\eta)^2+j(1-\eta)\eta^3) \\ \Pi_2^{\mathrm{MT}}(\alpha,\eta) &= \alpha\eta^2(k+m(1-\eta)+p(1-\eta)\alpha+q\alpha^3) \end{aligned} \tag{25}$$

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with fitted parameters:

$$\text{MT}: \begin{cases} g=-1.2078, & k=8.7145 \\ h=0.4907, & m=-40.6615 \\ j=-1.7257, & p=74.0617 \\ & q=-64.094 \end{cases}$$

- For the self-consistent scheme, which corresponds to a perfectly disordered porous material, the following expressions for $\Pi_1(\eta, \eta_0=1/2)$ and $\Pi_2(\alpha, \eta, \eta_0=1/2)$ are obtained:

$$\begin{aligned} \Pi_1^{\text{SC}}(\eta) &= \frac{\sqrt{2(2\eta-1)}-(2\eta-1)}{\sqrt{2}-1}(1+g(1-\eta)+h(1-\eta)^2+j(1-\eta)^3) \\ \Pi_2^{\text{SC}}(\alpha,\eta) &= \frac{2\eta-1}{2}(k+m(1-\eta)+p(1-\eta)\alpha+q\alpha^3) \end{aligned} \tag{26}$$

with fitted parameters:

$$\begin{aligned} g&=-5.3678, & k&=6.7374 \\ h&=12.1933, & m&=-39.5893 \\ j&=-10.3071, & p&=34.3216 \\ & & q&=-21.2053 \end{aligned} \tag{27}$$

With relationships between packing density, cohesive-frictional strength parameters, and indentation hardness in place and well approximated by smooth, closed-form fitting functions, an inverse approach to analysis of nanoindentation results is possible.

2.4. Inverse approach—packing densities and strength properties from nanoindentation tests

Nanoindentation is the only experimental method that can directly access the mechanical properties of the porous clay (Level 'I', see Figure 1) in shale. Bobko and Ulm [5] have demonstrated a detailed treatment of the issues relating to performing nanoindentation on shale, including an appropriate surface preparation procedure and relevant choices of indentation parameters in response to transverse isotropy and heterogeneity. The Bobko and Ulm data is employed in this study, along with results from indentation on a series of Woodford shales [8].

In general, force-driven nanoindentation tests operated to a maximum force of $P=4.8\,\text{mN}$ were carried out on all samples, leading to an average maximum indentation depth on the order of $h_{\max}=500$–$2000\,\text{nm}$ depending on the packing density of the porous clay phase. This indentation depth satisfies the scale separability conditions required for the indentation analysis based on continuum models, at the scale of Level 'I'. Shale samples were tested in the (macroscopically identified) direction of symmetry ($\underline{e}_3$) and normal to the direction of symmetry ($\underline{e}_1$). For each exposed surface, at least 300 tests were carried out on a regular grid of 30 μm grid spacing (distance between indents), which yields a sufficiently large number of tests over a wide area to achieve good sampling representation and statistical independence between each test.

With the derivations and analyses complete and in place, an inverse approach motivated by the one introduced by Ulm *et al.* [31] is implemented to estimate solid cohesive-frictional strength parameters. Nanoindentation assesses the mechanical behavior of the porous clay composite at Level 'I', via measured indentation moduli $M^{(1)}$ or $M^{(3)}$ (where the superscripts $(1,3)$ stand for indentation in the direction parallel to bedding and normal to bedding) and indentation hardness, H. The aim of the inverse approach is to translate the information about the porous clay composite at Level 'I' into information about the local packing density associated with each indentation test, as well as the solid, Level '0' strength properties of a particular sample.

We first demonstrate the feasibility of such an inverse analysis. Consider two typical arrays of indentations on a particular shale sample, assessing properties both normal and parallel to the axis

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of elastic symmetry. Together the two arrays consist of N individual tests. The properties of the solid phase (indentation moduli $m_s^{(1)}$ or $m_s^{(3)}$ and strength parameters c_s and α) are assumed to be constant for a given shale material. On the other hand, the locally observed packing density, η_i, where $i=1\ldots N$, may vary from indent to indent. The indentation moduli, $M_i^{(1)}$ or $M_i^{(3)}$, and indentation hardnesses, H_i, are composite properties that represent the homogenized (Level 'I') response of the porous materials (pores and Level '0').

In an inverse application, the unknowns of the problem are the properties of the solid phase, $m_s^{(1)}$ (or $m_s^{(3)}$), c_s, and α, and the N local packing densities, η_i, for a total of $N+4$ unknowns. The results of each indentation test, M_i and H_i, are known, for a total of $2N$ knowns. Consequently, for any large number of indentation tests, the system of equations is highly over-determined and an estimation of $m_s^{(J)}$, c_s, and α, and the N local packing densities, η_i should be feasible.

An algorithm developed and implemented in MATLAB works with the indents associated with the porous clay phase only. Although there may be variability in the mechanical properties of the solid phases, the scale separability conditions together with the central limit theorem produces characteristic quantities that are constant from one area of indentation to another, given that N is large. The inversion process appeared to achieve better convergence if the solid indentation modulus in each direction, $m_s^{(1)}$ or $m_s^{(3)}$, was constrained. Determination of $m_s^{(1)}$ and $m_s^{(3)}$ by average indentation analysis [5] is complemented by determination (and validation) of $m_s^{(1)}$ and $m_s^{(3)}$ by an inverse analysis of macroscopic elasticity measurements [2]. Computationally, the effect of this choice is to narrow the range of the possible local packing densities, η_i. In turn, this narrows the possible shapes of the H–η scaling relationships for each shale, and helps to optimize the estimation of the unique strength properties for each shale.

The algorithm is implemented as a minimization problem. The error is calculated as a 'normalized absolute error,' with the error on each point calculated as

$$S_i = \frac{(M_{i,\exp} - M_i^{(J)}(\eta_i, m_s^{(J)}))^2}{M_0^{(J)}} + \frac{(H_{i,\exp} - H_i(\eta_i, \alpha, c_s))^2}{H_0} \tag{28}$$

where $J=1,3$ for indentation parallel or normal to the elastic axis of symmetry, and where $M_0^{(J)}$ and H_0 are normalization factors, constant for the entire data set, typically chosen close to $m_s^{(1)}$, $m_s^{(3)}$, and h_s. Because each indentation test is seen as an independent statistical event, the minimization problem can be written as:

$$\min_{c_s,\alpha_s} \sum_i \min_{\eta_i} S_i(\eta_i, c_{s,} \alpha) \tag{29}$$

The minimization function is not convex, hence a testing procedure was used to select the best parameters, rather than a gradient algorithm.

3. RESULTS

3.1. Validation

The inverse approach is applied to a large database of indentations on various shales, including the data reported on in Bobko and Ulm [5] and Abousleiman *et al.* [8]. The inverse approach generates estimated packing density distributions, which give further insight into the microstructure of the material, and provide a means to validate the inverse approach.

Figure 5 shows typical plots of indentation modulus, M, and indentation hardness, H, versus the local clay packing density estimated for each individual indentation test on a single shale sample. The experimental points are randomly distributed on both sides of the model curves, which *a posteriori* justifies the scaling model presented above. The large range of possible packing densities, even after the considered tests were selected to be representative of the porous clay phase only, demonstrates the highly heterogeneous nature of the porous clay phase in shales.

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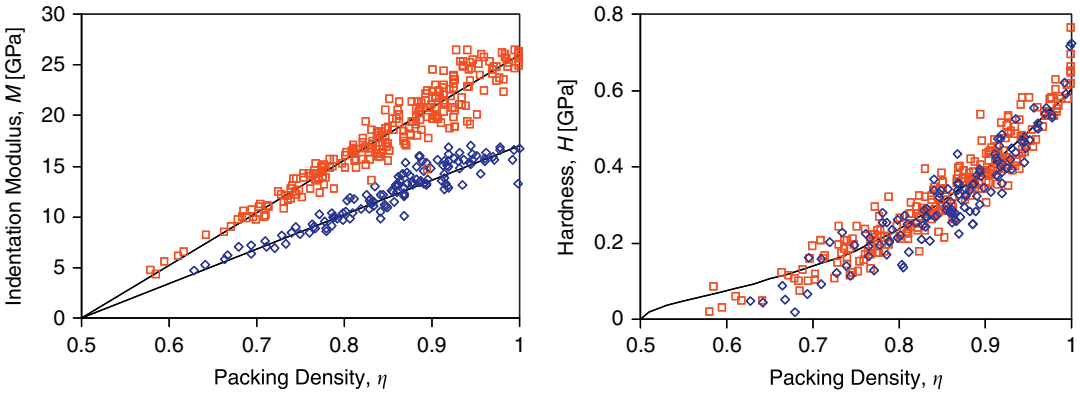


Figure 5. Scaling of individual nanoindentation test results. Indentation modulus (left) has an anisotropic scaling relationship with packing density, while hardness (right) has an isotropic scaling relationship with packing density.

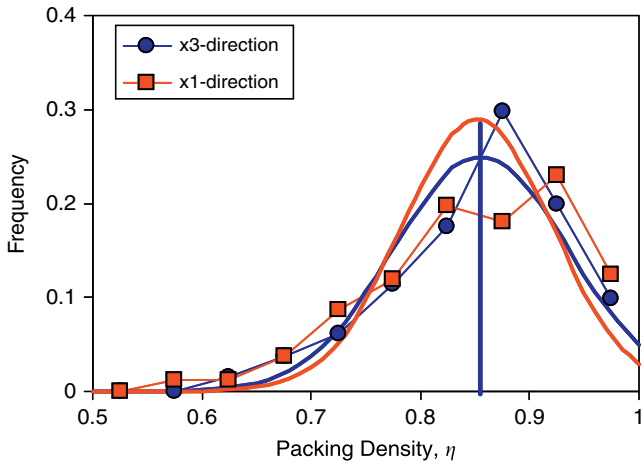


Figure 6. Packing density distributions for Shale 7, derived from indentation in the $x3$-direction and the $x1$-direction. The results overlap, suggesting that the shape of particles and pores have no privileged orientations. The mean value from indentation analysis of $\overline{\eta}=0.854$ matches well the mean value of $\overline{\eta}=0.875$ from mineralogy and porosity information.

The local packing densities associated with each individual indentation test can be summarized as statistical distributions of packing density. Figure 6 (top) displays frequency plots of packing densities in both the $x1$-direction and the $x3$-direction for Shale 7. The overlap between packing density distributions for each direction of testing is remarkable (as displayed in the example of Figure 6 and listed in Table I), and justifies *a posteriori* the choice of considering a perfectly disordered (spherical representation) mechanical morphology for the porous clay phase in shale. This observation implies that there is no orientation of pore space which would cause packing densities to have different values in different directions.

Table I displays the mean and standard deviation of packing densities estimated from each direction of testing for each tested shale, as well as an overall mean and standard deviation for each shale (with data from both directions of testing). The table also displays volumetric information about the mineralogy of the samples from [5, 8].

Comparing the overall mean of estimated packing densities with the average packing densities derived from mineralogy validates the inverse homogenization approach. Figure 7 compares packing density estimated by nanoindentation and mineralogy (based on Equation (1)). The results are well correlated, with a coefficient of determination, $r^2=0.81$. Moreover, there is no apparent trend

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Table I. Packing density and volume fraction information from indentation and from mineralogy for each tested shale.

	$\eta_{\text{indent,x}-1}$		$\eta_{\text{indent,x}-3}$		$\bar{\eta}_{\text{indent}}$		$\bar{\eta}_{\text{mineral}}$		$\overline{f}^{\text{inc}}$		$\overline{\phi}$	
Shale	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	+/−	Mean	+/−	Mean	+/−
1	0.674	0.081	0.655	0.079	0.666	0.010	0.668	0.021	0.159	0.004	0.279	0.019
2	0.770	0.082	0.743	0.107	0.760	0.012	0.804	0.022	0.230	0.005	0.151	0.018
3	0.890	0.062	0.890	0.080	0.871	0.010	0.896	0.001	0.275	0.001	0.075	0.000
4	0.775	0.090	0.782	0.111	0.771	0.012	0.691	0.016	0.326	0.005	0.209	0.012
7	0.856	0.080	0.852	0.091	0.854	0.013	0.875	0.022	0.274	0.005	0.091	0.017
Light	0.814	0.078	0.789	0.101	0.805	0.014	0.783	0.036	0.629	0.014	0.081	0.017
Dark	0.795	0.099	0.858	0.105	0.824	0.011	0.847	0.057	0.604	0.016	0.062	0.025
131	0.741	0.057	0.736	0.069	0.742	0.014	0.744	0.044	0.386	0.014	0.157	0.031
154	0.811	0.064	0.724	0.084	0.761	0.015	0.721	0.080	0.376	0.026	0.177	0.057
166	0.736	0.085	0.699	0.086	0.716	0.014	0.719	0.035	0.354	0.011	0.182	0.026
175	0.770	0.069	0.767	0.089	0.771	0.015	0.695	0.001	0.378	0.000	0.190	0.000
185	0.731	0.079	0.681	0.084	0.707	0.009	0.691	0.076	0.351	0.025	0.202	0.057

$\eta_{\text{indent,x1}}$ is the packing density from analysis of indentation tests in the $x1$-direction, $\eta_{\text{indent,x3}}$ is the packing density from analysis of indentation tests in the $x3$-direction, and $\bar{\eta}_{\text{indent}}$ is the packing density from analysis of indentation tests in both directions. $\bar{\eta}_{\text{mineral}}$ is the average packing density calculated from mineralogy and porosity, $\bar{f}^{\text{inc}}$ is the average inclusion volume fraction calculated from mineralogy and porosity, and $\bar{\phi}$ is the average porosity calculated from mercury intrusion porosimetry and bulk density information. For packing densities from indentation, the standard deviation (S.D.) is that of the distribution of packing densities. For mineralogy data, the variation (+/−) is a result of various methods of obtaining volume fractions; for more information, see [5].

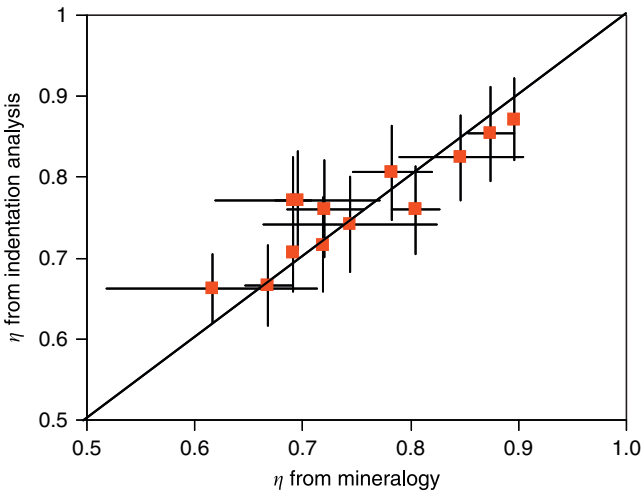


Figure 7. Comparison of packing densities derived from mineralogy estimates and from nanoindentation analysis. Displayed uncertainties in mineralogy estimates come from different measurement techniques. Displayed uncertainties in indentation estimates are representative of two standard deviations.

in the difference between techniques of measuring average packing density. The experimental observations provide little evidence of a compaction or consolidation effect in the nanoindentation experiments and analysis.

It is tempting to put a length scale with the characteristic packing of the porous clay phase. This is achieved here by considering pore throat radius distributions determined from mercury intrusion porosimetry on the same materials (Figure 8). Mercury intrusion porosimetry [46] is a popular technique for measuring pore size distributions, although the experimental and analytical issues with mercury intrusion porosimetry are well known [47]. It has been shown, however, that for

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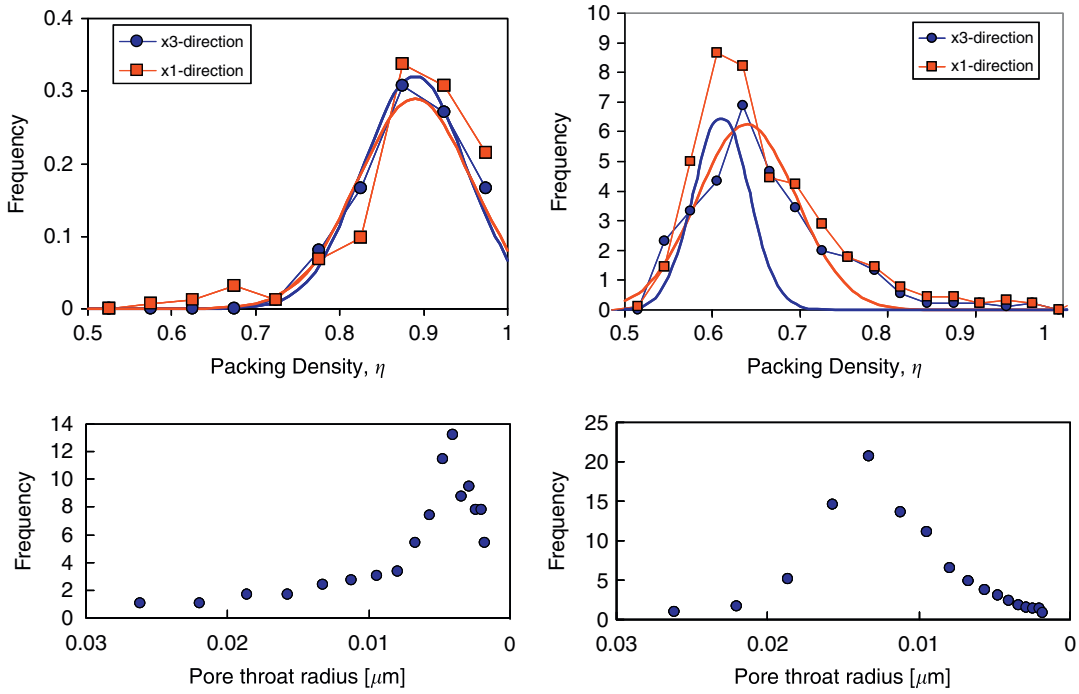


Figure 8. Link between packing density distributions (top) and pore size distributions (bottom). Data from Shale 3 (left) and Shale 1 (right). Mercury porosimetry data courtesy of Chevron.

Table II. Mean pore throat radius, r, calculated from pore throat size distributions, with mean and standard deviation (S.D.) of packing density from indentation analysis, $\bar{\eta}_{\text{indent}}$, duplicated from Table I. Mercury porosimetry tests performed by Chevron.

		$\bar{\eta}_{\text{indent}}$	
Shale	r (nm)	Mean	S.D.
1	12.46	0.666	0.010
2	4.56	0.760	0.012
3	3.37	0.871	0.010
4	8.32	0.771	0.012
Dark	4.22	0.824	0.011
131	11.21	0.742	0.014
154	10.65	0.761	0.015
166	11.1	0.716	0.014
175	9.33	0.771	0.015
185	11.27	0.707	0.009

similarly structured geomaterials, comparisons between pore-size distributions are reliable [48]. Table II displays the mean and standard deviation of pore throat radius estimated from mercury intrusion porosimetry for the tested shales. Figure 9 compares the mean pore throat radius with the average packing density from nanoindentation analysis. The average packing density tends to decrease as the average pore throat radius increases.

3.2. *Cohesive frictional strength parameters*

The estimated strength parameter results are shown in Table III.

3.2.1. Invariant solid contact hardness. The Level '0' contact hardness, h_s, is recovered when a clay packing density of $\eta = 1$ is input into the Level 'I' scaling relationship for hardness (22).

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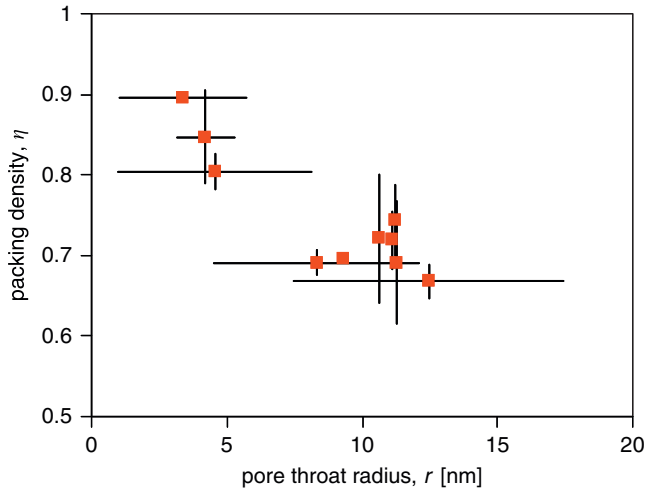


Figure 9. Comparison of average pore throat radius obtained from mercury intrusion porosimetry and average packing density obtained from nanoindentation. Uncertainties represent two standard deviations.

Table III. Particle hardness h_s, friction coefficient, α and cohesion, c_s for a Drucker–Prager solid, for each of the tested shales.

	h_s (GPa)		α		c_s (GPa)	
Shale	Mean	S.D.	Mean	S.D.	Mean	S.D.
1	0.588	0.103	0.408	0.116	0.044	0.012
2	0.620	0.067	0.218	0.058	0.078	0.014
3	0.781	0.069	0.057	0.036	0.143	0.018
4	0.564	0.053	0.152	0.036	0.083	0.006
7	0.567	0.059	0.073	0.038	0.100	0.038
Light	0.699	0.065	0.205	0.114	0.090	0.007
Dark	0.428	0.039	0.108	0.027	0.070	0.005
131	1.104	0.085	0.141	0.035	0.166	0.020
154	0.802	0.079	0.160	0.050	0.115	0.013
166	0.720	0.088	0.380	0.060	0.059	0.006
175	0.772	0.082	0.093	0.084	0.129	0.039
185	0.755	0.055	0.292	0.029	0.078	0.006

As shown in Figure 10, the contact hardness and the average clay packing density are uncorrelated. In fact, the porous clay phase in the shale samples, despite possessing wide variations in clay mineralogy, has a relatively constant contact hardness of $h_s = 0.69 \pm 0.09$ GPa. The contact hardness is identified as an intrinsic strength parameter that characterizes the strength behavior of shale at Level '0.'

3.2.2. Scaling of particle friction and cohesion with packing density. In contrast to the contact hardness, the Drucker–Prager friction coefficient, α, varies strongly from shale to shale, as a function of the average clay packing density as shown in Figure 11 (top). The friction coefficient decreases as the average clay packing density increases, and appears to tend toward zero as the clay packing density increases to one. Note that the experimentally obtained friction coefficient, α, for any of the considered shale samples does not exceed the computational convergence limit of $\alpha > 1/\sqrt{3}$, even though this constraint was not placed on the optimization.

Figure 11 (bottom) shows a direct relationship between cohesion and average clay packing density. As the average packing density increases, the cohesion increases as well. The cohesion appears to tend toward zero as the clay packing density goes to the percolation threshold of 0.5.

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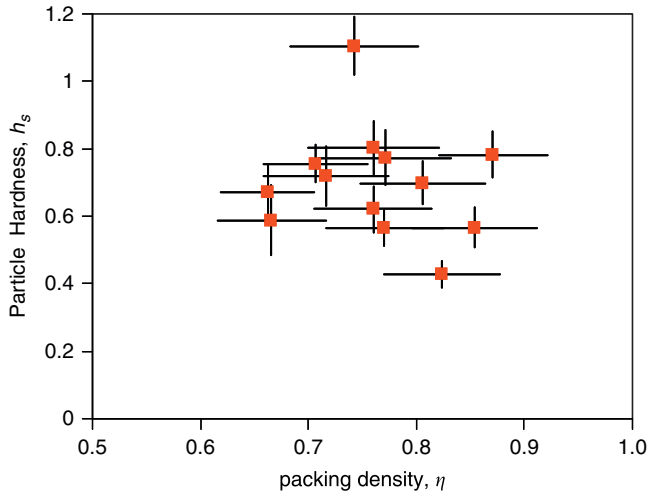


Figure 10. Particle hardness shows no apparent trend with packing density. Instead particle hardness is a roughly constant quantity. Uncertainties represent two standard deviations.

There is somewhat more scatter in the cohesion/packing density relationship than there is in the friction/packing density relationship. In fact, the biggest 'outlier' is the same as that in the contact hardness analysis. The source of this behavior is unclear, but does not affect the overall trends.

4. DISCUSSION

The two interesting observations that emerge from the application of the inverse approach to shale nanoindentation is (1) an almost constant particle hardness for all types of shales of different mineralogy and porosity; and (2) a dependence of the Drucker–Prager model parameters (solid cohesion c_s and friction coefficient α) on the mean packing density of the porous clay phase. The focus of the discussion is to reconcile these two observations.

4.1. Cohesion and coordination number, bond distance

The observation that cohesion c_s increases with mean packing density deserves investigating other morphological parameters linked with packing density. Consider the coordination number, the number of contacts between particles, which increases with the packing of particles. Oda [49] has determined the mean coordination numbers of randomly packed monosize and two-sized spherical assemblies. These results, as well as theoretical packings of spheres, are recalculated in terms of packing density and are displayed in Figure 12: The coordination number increases almost linearly with packing density, which corresponds well to the trend of cohesion and packing density displayed in Figure 11 (bottom). In fact, for a random assembly of cemented spheres, Ingles [50] has derived an estimate of the cohesion of the system that varies linearly with the coordination number, as well as on the relative amount of cement and the average bond strength per contact zone. This means that the evolution of the mean solid cohesion with packing density may well be attributable to the mean number of contacts between particles which is the higher the more packed the system.

4.2. The origin of static friction

Things are somewhat more complex when it comes to friction. Indeed, researchers have long sought to understand the physical mechanisms responsible for friction between solids, and the physical origin of friction is still a thriving research topic [51–55], particularly in granular materials [12, 13, 56]. The results presented here add some compelling new evidence to the discussion,

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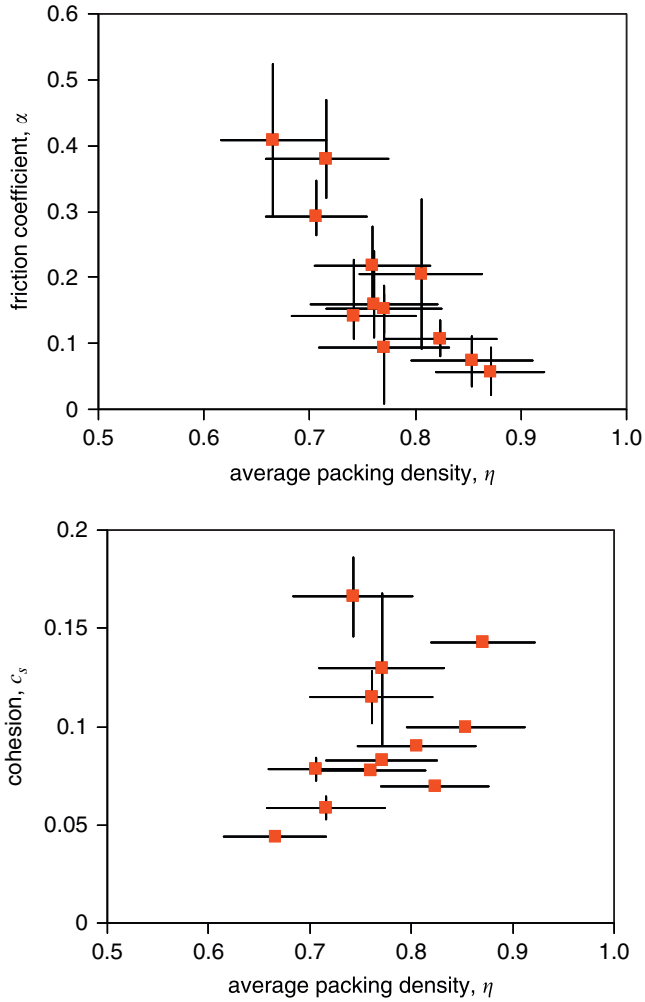


Figure 11. Scaling of clay strength properties with packing density. Top: friction coefficient, Bottom: cohesion. Uncertainties represent two standard deviations.

wherein the friction between particles decreases with the packing density; and a solid without pores has no friction.

Here, a comment on the model assumptions is in order. Frictional behavior is classically described by Amonton's laws; the surface friction force increases linearly with the load pushing two objects together, while remaining independent of the surface area of the contact. This is captured in strength of materials by the Mohr–Coulomb strength criterion, a surface traction strength criterion:

$$f(\underline{T}=\boldsymbol{\sigma}\cdot\underline{n})=|T_t|+\mu\ T_n-C\leqslant 0 \tag{30}$$

with $T_t=\underline{t}\cdot(\boldsymbol{\sigma}\cdot\underline{n})$ and $T_n=\underline{n}\cdot(\boldsymbol{\sigma}\cdot\underline{n})$ the tangential shear and normal stresses on the material surface oriented by unit outward normal $\underline{n}$. The model we here use for the solid is a Drucker–Prager criterion, which in contrast to the Mohr–Coulomb criterion is an isotropic cohesive-frictional strength criterion

$$f(\boldsymbol{\sigma})=\sqrt{J_2}+\alpha\sigma_m-c\leqslant 0 \tag{31}$$

where $\sqrt{J_2}=\sqrt{\frac{1}{2}s_{ij}s_{ij}}$ and $\sigma_m=\frac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})$ represent tangential shear and normal stresses built in stress space on a fictitious plane (the deviator plane) oriented by the hydrostatic stress axis. From a heuristic point of view the Mohr–Coulomb strength parameters (cohesion C, friction coefficient

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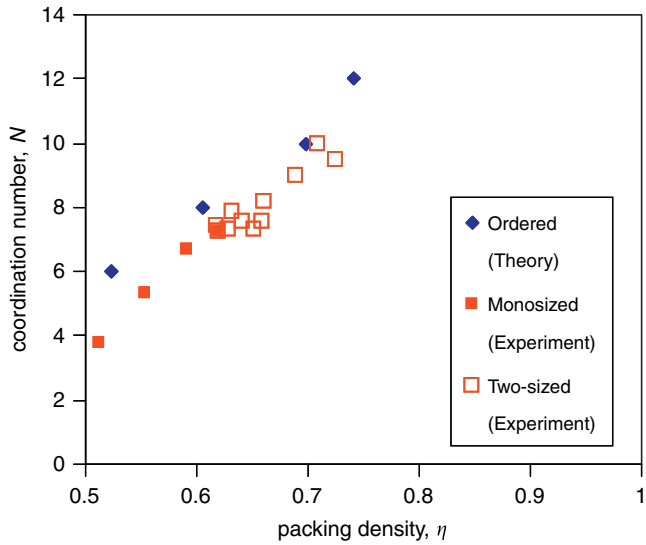


Figure 12. Mean coordination numbers determined for theoretical packings of particles and for experiments on random packings of monosized and two-sized glass beads. For experimental values, the mean packing densities are reported. Mean coordination numbers increase roughly linearly with packing density. Data from [49], replotted in terms of packing density.

$\mu = \tan\varphi$) can be related to the Drucker–Prager strength parameters (cohesion c, friction coefficient α) by assuming same strength values for specific stress states [57]:

$$\begin{aligned} &\text{Compression Cone:} \quad C = \frac{3-\sin\varphi}{2\sqrt{3}\cos\varphi}c, \quad \sin\varphi = \frac{3\alpha}{\alpha+2\sqrt{3}} \\ &\text{Internal Cone:} \quad C = \sqrt{\frac{3+(\sin\varphi)^2}{3(\cos\varphi)^2}}, \quad \sin\varphi = \sqrt{\frac{3\alpha^2}{3-\alpha^2}} \end{aligned} \tag{32}$$

From a more fundamental perspective, recent results of micromechanics show that the continuum Drucker–Prager criterion can represent an averaged form of a Mohr–Coulomb criterion (30) on the interfaces between linear elastic rigid polycrystal particles [20]. It is in this sense that we employ the Drucker–Prager strength model to describe the strength domain of the agglomerated polycrystal clay particles at level '0'.

With this in mind, it is a useful reminder that frictional forces have been classically explained as generated by the interlocking of surface asperities [58]. For one surface to slide past another, it must be lifted up and over a surface asperity of some slope. An energy argument against this model was presented as early as 1804 by Leslie [51]. Surfaces with asperities that are both periodic and aligned fit the geometrical model well. But for randomly rough surfaces, upwards sloping asperities are balanced by downwards sloping asperities. In this case, frictional force vanishes. A theory which recognizes that material surfaces are covered by adsorbed molecules (i.e. hydrocarbons, water molecules, etc.) has been proposed as a solution to this problem [54]. Introducing enough of these small, mobile molecules into the interface yields a frictional coefficient that obeys Amonton's laws. Another new modeling approach to the origin of friction has been presented [52]. This approach suggests that a propagation of self-healing cracks, similar to sliding a rug across the floor by moving a bump or the movement of dislocations in crystalline materials, is a micro-scale mechanism which leads to the results described by Amonton's laws.

With the models of Müser *et al.* [54] and Gerde and Marder [52] in mind, the relationship observed in Figure 11 (top) may be better explained by considering the interactions between particles and pores. For adsorbed molecules to be present and mobile between surfaces, some space must exist between particles. Similarly, some space between particles must be present to permit a

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self-healing microcrack to form. A link was established between the average clay packing density and pore throat radius determined by mercury porosimetry. Given the correlation between porosity and pore throat radius on the one side (Figure 8), and the correlation between friction coefficient and packing density (Figure 11 (top)) on the other, it is expected that the friction coefficient increases as the mean pore throat radius increases. Loose packings of particles apparently permit larger pore spaces, which in turn permit the possibility of larger asperities under a conventional frictional model, more space for mobile molecules in the Müser *et al.* model, or greater ability for microcracks to form in the Gerde and Marder model. The result in any case is that the friction coefficient is increased when the spaces between particles are larger. Conversely, in the limit of zero porosity, or infinitely small pore throat radius, the problem is reduced to particles without asperities, bare incommensurate surfaces, or inhibition of microcracks; the friction coefficient goes to zero.

4.3. Fractal scaling of friction

The scaling of the friction coefficient with the porosity respectively with the pore throat radius deserves some further analysis. To this end, we first translate the Drucker–Prager friction coefficient into the Mohr–Coulomb friction coefficient, via the internal cone equivalence (32) such that $\alpha \to \mu = \tan\varphi$.

Then, recall that Amontons' Laws of friction state that the frictional force F_t is proportional to the normal force F_n holding the two surfaces together (i.e. $\mu \sim F_t/F_n$), and that this force is independent of the area of contact. In this frictional contact problem between spheres there are two lengths of different magnitudes: the length in the normal direction is defined by the particle radius, ε, while the length in the tangential direction is defined by the contact radius, a, which is typically much smaller than the sphere radius; $a \ll \varepsilon$. This different order in magnitude motivates the consideration of an extended base dimension system for dimensional analysis, in which:

$$[F_t] = L_t M T^{-2}, \quad [F_n] = L_n M T^{-2}, \quad [\mu] = L_t/L_n \tag{33}$$

with L_t representing tangential length base dimensions in the plane of contact of the two surfaces ($[a] = L_t$), and L_n the length base dimension measured in the normal direction ($[\varepsilon] = L_n$). For instance, the perimeter s of a sphere is measured along any circle of radius ε, and has thus dimension $[s] = [\varepsilon] = L_n$. In return, the surface area p of a sphere corresponds exactly to the tangential area measurement with dimension; $[p] = L_t^2$. Following this analysis, the dimension function of the friction coefficient is:

$$[\mu] = L_t/L_n = [p]^{1/2}[s]^{-1} \tag{34}$$

It is now interesting to combine these dimensional considerations with some basic elements of fractal packing; cf. [59, 60]. In such a fractal packing, the perimeter is a line fractal, while the surface area is a surface fractal, defined respectively by the asymptotic relations $s \sim \varepsilon^{1-D}$ and $p \sim \varepsilon^{2-D}$, with D the fractal dimension of the packing. A volume fractal scaling relation links either solid or pore volume with a radius by

$$\varphi \sim \varepsilon^{3-D} \tag{35}$$

which is a fractal scaling seen in many natural materials [61].

Combining the fractal packing relations with the dimensional analysis of Amonton's Laws suggests that the friction coefficient should obey an asymptotic scaling of the form:

$$\mu \sim \sqrt{\varepsilon^{2-D}}(\varepsilon^{1-D})^{-1} = \varepsilon^{\frac{D}{2}} \tag{36}$$

Finally, substituting clay porosity φ for particle radius ε using the volume fractal scaling relation (35), shows that a fractal scaling power relation of the form

$$\mu \sim (\varphi^{\frac{1}{3-D}})^{\frac{D}{2}} = \varphi^{\gamma}, \quad D = \frac{6\gamma}{2\gamma+1} \tag{37}$$

links the friction coefficient to porosity.

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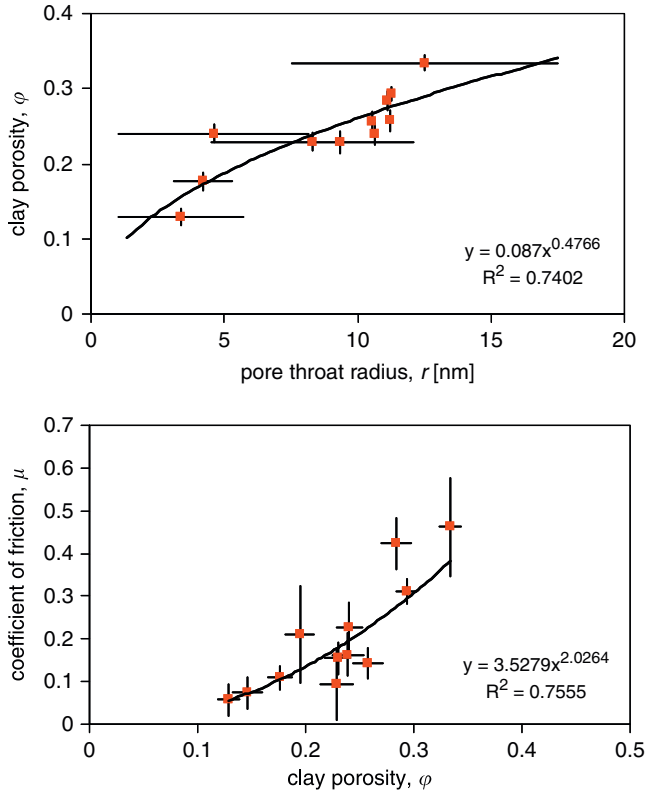


Figure 13. Fractal scalings of clay packings. Top: Mean clay porosity φ, determined from nanoindentation scaling analysis versus pore throat radius, r, from poromercury intrusion. Bottom: Scaling of Mohr–Coulomb friction coefficient μ versus clay porosity φ. In both figures, the power law fits are given by the displayed equation. Uncertainties in experimental data represent plus and minus one standard deviation.

Figure 13 (top) shows that the packing density of different materials (with different mineralogy and porosity) scales with the pore throat radius, and a power law fitting gives $\varphi \sim l^{0.477}$ ($r^2 = 0.7402$). Using (35), the fractal dimension is $D = 2.523$. Figure 13 (bottom) shows the scaling of the friction coefficient with the porosity; where a power law fitting gives $\mu \sim \varphi^{2.0264}$($r^2 = 0.7555$). From the experimentally fitted power exponent of the friction–porosity relation and (37), a fractal dimension $D = 2.406$ is obtained.

These fractal dimensions are very similar, and bracket the fractal dimension of a perfect space-filling bearing, known as the Apollonian packing, for which $D = 2.474$ [60]. One of the important features of space-filling bearings is that particles rotate without slip and without torsion, which minimizes energy dissipation due to friction [62]. Thus, space-filling bearings should exhibit a negligible pressure dependency in the sense of Amontons' laws of friction. In contrast to space-filling bearings that achieve an asymptotic packing density of one, real nanogranular materials like the porous clay composite in shale never achieve this asymptotic packing. We thus find that the well-known pressure sensitivity of these materials, expressed by inter-particle friction, is representative of an 'imperfect' space-filling bearing. The imperfections are related to porosity; that is, the material becomes frictional due to the presence of porosity, while it is frictionless for a solid without pores.

5. CONCLUSIONS

The fundamental idea of the approach developed here is that it is possible to assess porous nano-structure and nanomechanical properties from the implementation of novel strength homogenization

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models for the indentation analysis of a large array of nanoindentation tests on heterogeneous materials like compacted clay in shale:

(1) In contrast to the pronounced anisotropy of shale elasticity at multiple scales [27, 34], packing density distributions and strength properties are found to be isotropic, suggesting that particle shape does not play a significant role in the limit strength behavior of the porous clay in shale. This conclusion pertains to an 'intact' rock sample and does not extend to shales with fractures and cracks. As such the isotropic strength behavior may be viewed as a baseline behavior for shale strength behavior, such that well-known anisotropic features in the macroscopic strength behavior of shale can be attributed with confidence to cracks, fractures, and other geologic features such as joints and faults.
(2) The well-known pressure sensitivity of shale, expressed by inter-particle friction, is related to porosity; that is, the material exhibits friction due to the presence of porosity, and is frictionless for a solid without pores. This observation of a porosity induced inter-particle friction can be viewed as an extension of molecular dynamics simulation results, which attributed the prevalence of static friction to the presence of a 'third body' of adsorbed molecules (hydrocarbons, water molecules etc.) that arrange to lock two contacting surfaces together [53]. Moreover, the observation that this inter-granular friction exhibits a pronounced fractality can be seen as a consequence of a highly disperse size distribution of grains, similar to space-filling bearings as the fractal dimension indicates. Yet, in contrast to a space-filling bearing which is such that a perfect packing allows achieving an asymptotic packing density of one, compacted clay in shale never achieves this asymptotic packing, due to the presence of a 'third body' filling the porosity, from the scale of some molecules adsorbed at surfaces to nanometer scales. Thus it is the nanogranular nature of these materials and its distinct porosity which makes shales distinct frictional.
(3) As inter-particle friction decreases, the solid cohesion increases with the packing; a phenomenon that may be attributable to the number of contacts stabilizing particles within the compacted clay. This explanation is consistent with the fractal scaling model evoked here above, if one associates a characteristic coordination number with each particle, while noting that the number of particles scales asymptotically according to $N \sim \varepsilon^{D}$; that is with the porosity $N \sim \varphi^{-D/(3-D)}$. Thus, with a fractal dimension, $D<3$, the number of contact points increases with decreasing porosity or increasing packing.
(4) The two opposed scaling phenomena appear to compensate each other such that the particle-to-particle hardness is, in first order, constant, $h_s = 0.69 \pm 0.09\,\text{GPa}$, for a wide variety of shales of different mineralogy, packing, and probed in different directions.

The overall picture which thus emerges is that compacted clay in shale possesses a distinct nanogranular nature that drives its nanoscale cohesive-frictional properties. It is expected that the implementation of this nanogranular signature of shale in strength upscaling models will lead to the development of predictive engineering models for geomechanics field-characterization of shale in high demand by the industry [63].

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REFERENCES

1. Thomsen L. Seismic anisotropy. *Geophysics* 2001; **66**:40–41.
2. Ortega JA, Ulm F-J, Abousleiman Y. The effect of the nanogranular nature of shale on their poroelastic properties. *Acta Geotechnica* 2007; **2**:155–182.
3. Lee SY, Hyder LK, Alley PD. Microstructural and mineralogical characterization of selected shales in support of nuclear waste repository studies. In *Microstructure of Fine-grained Sediments, from Mud to Shale*, Bennet RH, Bryant WR, Hulbert MH (eds). Springer: New York, 1991; 545–560.

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4. Ulm F-J, Delafargue A, Constantinides G. Experimental microporomechanics. In *Applied Micromechanics of Porous Materials*, Ulm F-J, Dormieux L (eds). CISM Lecture Notes, No. 580. Springer: New York, 2005; 207–288.
5. Bobko C, Ulm F-J. The nano-mechanical morphology of shale. *Mechanics of Materials* 2008; **40**:318–337.
6. Jizba D. Mechanical and acoustical properties of sandstones and shales. *Ph.D. Dissertation*, Stanford University, Palo Alto, 1991.
7. Horsrud P, Sonstebo EF, Boe R. Mechanical and petrophysical properties of North Sea shales. *International Journal of Rock Mechanics and Mining Sciences and Geomechanics Abstracts* 1998; **35**(12):1009–1020.
8. Abousleiman Y, Tran M, Hoang S, Bobko C, Ortega A, Ulm F-J. Geomechanics field and lab characterization of Woodford Shale: the next gas play. *2007 Annual SPE Technical Conference and Exhibition*, Anaheim, CA, 11–14 November 2007; SPE 110120.
9. Colmenares LB, Zoback MD. A statistical evaluation of intact rock failure criteria constrained by polyaxial test data for five different rocks. *International Journal of Rock Mechanics and Mining Science* 2002; **39**:695–729.
10. Schofield AN, Wroth P. *Critical State Soil Mechanics*. McGraw-Hill: New York, 1968.
11. Wood DM. *Soil Behavior and Critical State Soil Mechanics*. Cambridge University Press: Cambridge, U.K., 1990.
12. Kruyt NP. Contact forces in anisotropic frictional granular materials. *International Journal of Solids and Structures* 2003; **40**:3537–3556.
13. Roux J-N. Geometric origin of mechanical properties of granular materials. *Physical Review E* 2001; **61**(6): 6802–6836.
14. Gurson AL. Continuum theory of ductile rupture by void nucleation and growth. Part I. Yield criteria and flow rules for porous ductile media. *Journal of Engineering*: *Materials and Technology*, *Transactions on ASME* 1977; **99**:2–15.
15. Gologanu M, Leblond J-B, Perrin G, Devaux J. Recent extensions of Gurson's model for porous ductile metals. In *Continuum Micromechanics*, Suquet P (ed.). Springer: New York, 1997; 61–130.
16. Barthélémy J-F, Dormieux L. Determination of the macroscopic strength criterion of a porous medium by nonlinear homogenization. *Comptes Rendus Mecanique* 2003; **331**(4):271–276.
17. Sanahuja J, Dormieux L. Strength of a porous medium with a heterogeneous solid phase. *Comptes Rendus Mecanique* 2005; **333**(11):818–823.
18. Trillat M, Pastor J. Limit analysis and Gurson's model. *European Journal of Mechanics and Solids* 2005; **24**:800–819.
19. Dormieux L, Kondo D, Ulm F-J. *Microporomechanics*. Wiley: New York, 2006.
20. Fritsch A, Dormieux L, Hellmich C, Sanahuja J. Micromechanics of crystal interfaces in polycrystalline solid phases of porous media: fundamentals and application to strength of hydroxyapatite biomaterials. *Journal of Material Science* 2007; **42**:8824–8837.
21. Cariou S, Ulm F-J, Dormieux L. Hardness–packing density scaling relations for cohesive-frictional porous materials. *Journal of the Mechanics and Physics of Solids* 2008; **56**:924–952.
22. Thore P, Pastor F, Pastor J, Kondo D. Closed-form solutions for the hollow sphere model with Coulomb and Drucker-Prager materials under isotropic loadings. *Comptes Rendus Mecanique* 2009; **337**(5):260–267.
23. Grim RE. The history of the development of clay mineralogy. *Clay Minerals* 1988; **36**(2):97–101.
24. Mitchell J, Soga K. *Fundamentals of Soil Behavior* (3rd edn). Wiley: New York, 2005.
25. Guggenheim S, Chollacoop N, Dao M, Li J, Minar AM, Shen Y-L. Summary of recommendations of nomenclature committees relevant to clay mineralogy: report of the Association Internationale Pour L'Etude des Argiles (AIPEA) Nomenclature Committee for 2006. *Clays and Clay Minerals* 2007; **54**(6):761–772.
26. Mavko G, Mukerji T, Dvorkin J. *The Rock Physics Handbook*. Cambridge University Press: Cambridge, U.K., 1998.
27. Ortega JA, Ulm F-J, Abousleiman Y. The nanogranular acoustic signature of shale. *Geophysics* 2009; **74**(3): D65–D84.
28. Ortega JA, Gathier B, Ulm F-J. Homogenization of cohesive-frictional strength properties of porous composites: linear comparison composite approach. *ASCE Journal of Nanomechanics and Micromechanics* 2010; in press.
29. Borges LA, Zouain N, Huespe AE. A nonlinear optimization procedure for limit analysis. *European Journal of Mechanics*, *A/Solids* 1996; **15**(3):487–512.
30. Pontes I, Borges LA, Zouain N, Lopes FR. An approach to limit analysis with cone-shaped yield surfaces. *International Journal for Numerical Methods in Engineering* 1997; **40**:4011–4032.
31. Ulm F-J, Vandamme M, Bobko C, Ortega JA, Tai K, Ortiz C. Statistical indentation techniques for hydrated nanocomposites: cement, bone, and shale. *Journal of the American Ceramic Society* 2007; **90**(9):2677–2692.
32. Oliver WC, Pharr GM. An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments. *Journal of Materials Research* 1992; **7**(6):1564–1583.
33. Zaoui A. Continuum micromechanics: Survey. *Journal of Engineering Mechanics* (ASCE) 2002; **128**(8):808–816.
34. Ortega JA, Ulm F-J, Abousleiman Y. The effect of particle shape and grain-scale properties of shale: a micromechanics approach. *International Journal for Numerical and Analytical Methods in Geomechanics* 2010; **34**:1124–1156. DOI: 10.1002/nag.850.
35. Delafargue A, Ulm F-J. Explicit approximations of the indentation modulus of elastically orthotropic solids for conical indenters. *International Journal of Solids and Structures* 2004; **41**:7351–7360.

Copyright © 2010 John Wiley & Sons, Ltd.

DOI: 10.1002/nag

36. Ponte Castañeda P. New variational principles in plasticity and their application to composite materials. *Journal of the Mechanics and Physics of Solids* 1992; **40**:1757–1788.
37. Ponte Castañeda P. Second-order homogenization estimates for nonlinear composites incorporating field fluctuations. I. Theory. *Journal of the Mechanics and Physics of Solids* 2002; **50**:737–757.
38. Constantinides G, Ulm F-J. The nanogranular nature of C-S-H. *Journal of the Mechanics and Physics of Solids* 2007; **55**(1):64–90.
39. Onoda GY, Liniger EG. Random loose packings of uniform spheres and the dilatancy onset. *Physical Review Letters* 1990; **64**:2727–2730.
40. Desrues J. Limitations du choix de l'angle de frottement pour le critère de plasticité de Drucker-Prager. *Revue Française de Génie Civil* 2002; **6**:853–862.
41. Brinell JA. Memoire sur les epreuves a bille en acier. *Congres International des Methodes d'Essai des Materiaux de Construction*, Paris, vol. 2, 1901; 83–94.
42. Chitkara NR, Butt MA. Numerical construction of axisymmetric slip-line fields for indentation of thick blocks by rigid conical indenters and friction at the tool–metal interface. *International Journal of Mechanical Sciences* 1992; **34**(11):849–862.
43. Locket FJ. Indentation of a rigid/plastic material by a conical indenter. *Journal of the Mechanics and Physics of Solids* 1963; **11**:345.
44. Tabor D. A simple theory of static and dynamic hardness. *Proceedings of the Royal Society London*, *Series A* 1948; **192**:247.
45. Ganneau FP, Constantinides G, Ulm F-J. Dual-indentation technique for the assessment of strength properties of cohesive-frictional materials. *International Journal of Solids and Structures* 2006; **43**(6):1727–1745.
46. ASTM D-4404-84. *Standard Test Method for Determination of Pore Volume and Pore Volume Distribution of Soil and Rock by Mercury Intrusion Porosimetry*. ASTM International: West Conshohocken, PA, 2004.
47. Diamond S. Mercury porosimetry: an inappropriate method for the measurement of pore size distributions in cement-based materials. *Cement and Concrete Research* 2000; **30**:1517–1525.
48. Radlinski AP, Ioannidis MA, Hinde AL, Hainbuchner M, Baron M, Rauch H, Kline SR. Angstrom-to-millimeter characterization of sedimentary rock microstructure. *Journal of Colloid and Interface Science* 2004; **274**:607–612.
49. Oda M. Co-ordination number and its relation to the shear strength of granular material. *Soils and Foundations* 1977; **17**(2):29–42.
50. Ingles OG. Bonding forces in soils—part 3. *Proceedings of the First Conference of the Australian Road Research Board*, vol. 1, 1962; 1025–1047.
51. Gao J, Luedtke WD, Gourdon D, Ruths M, Israelachvili JN, Landman U. Frictional forces and Amontons' law: from the molecular to the macroscopic scale. *Journal of Physical Chemistry B* 2004; **108**:3410–3425.
52. Gerde E, Marder M. Friction and fracture. *Nature* 2001; **413**:285–288.
53. He G, Muser MH, Robbins MO. Adsorbed layers and the origin of static friction. *Science* 1999; **284**(5420): 1650–1652.
54. Müser MH, Wenning L, Robbins MO. Simple microscopic theory of Amonton's laws for static friction. *Physical Review Letters* 2001; **86**(7):1295–1298.
55. Urback M, Klafter J, Gourdon D, Israelachvili J. The nonlinear nature of friction. *Nature* 2004; **430**:525–528.
56. Lemaitre A. Origin of a repose angle: kinetics of rearrangement for granular materials. *Physical Review Letters* 2002; **89**(6):064303.
57. Schwiger HF. On the use of Drucker–Prager failure criteria for earth pressure problems. *Computer and Geotechnics* 1994; **16**:223–246.
58. Bowden FP, Tabor D. *The Friction and Lubrication of Solids. Oxford Classic Texts in the Physical Sciences*. Clarendon Press: Oxford, U.K., 2001 (c1954).
59. Herrmann HJ, Mantica G, Bessis D. Space-filling bearings. *Physical Review Letters* 1990; **65**(26):3223–3226.
60. Borkovec M, De Paris W, Peikert R. The fractal dimension of the apollonian sphere packing. *Fractals* 1994; **2**(4):521–526.
61. Katz AJ, Thompson AH. Fractal sandstone pores: implications for conductivity and pore formation. *Physics Review Letters* 1985; **54**(12):1325–1328.
62. Mahmoodi Baram R, Herrmann HJ. Space-filling bearings in three dimensions. *Physics Review Letters* 2003; **92**(4):044301.
63. Abousleiman Y, Tran M, Hoang S, Ortega A, Ulm F-J. Geomechanics field characterization of Woodford Shale and Barnett Shale with advanced logging tools and nano-indentation on drill cuttings. *Leading Edge* 2010; in press.

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